6 b output.sci

```
        y_impulse(i) = b * exp(-a * t(i));
    end
end

// Step response
for i = 1:length(t)
    if t(i) >= 0 then
        y_step(i) = (b/a) * (1 - exp(-a * t(i)));
    end
end

// Plot all four signals
scf(0);
clf();

subplot(4,1,1);
plot(t, x_impulse);
xtitle("Input Signal - Impulse", "Time (t)", "x(t)");

subplot(4,1,2);
plot(t, y_impulse);
xtitle("Output Signal for Impulse Input", "Time (t)", "y(t)");

subplot(4,1,3);
plot(t, x_step);
xtitle("Input Signal - Step", "Time (t)", "x(t)");

subplot(4,1,4);
plot(t, y_step);
xtitle("Output Signal for Step Input", "Time (t)", "y(t)");

```

